C Interview Questions with Answers

1. What is C language?

C is a general-purpose, procedural programming language used for system and application programming.

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2. Who developed C?

C was developed by Dennis Ritchie at Bell Labs in 1972.

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3. What are the features of C?

- Simple
- Fast
- Portable
- Structured
- Supports pointers
- Efficient memory management

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4. What is the structure of a C program?

- Header files
- Global declarations
- main() function
- User-defined functions

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5. What is the purpose of main() function?

It is the entry point of a C program. Execution starts from main().

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6. What are header files?

Files containing function declarations and macros, such as stdio.h.

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7. Why do we use #include?

To include header files in a program.

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8. What are variables?

Named memory locations used to store data.

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9. What are data types in C?

They define the type of data a variable can store, such as int, float, and char.

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10. Difference between int, float, double, and char?

Type	Purpose
int	Integer values
float	Decimal values
double	Large decimal values
char	Single character

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11. What are constants?

Fixed values that cannot be changed during program execution.

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12. What are identifiers?

Names given to variables, functions, arrays, etc.

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13. What are keywords in C?

Reserved words with predefined meanings, such as int, if, and return.

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14. What is the use of sizeof() operator?

Returns the size of a variable or data type in bytes.

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15. What are arithmetic operators?

+, -, *, /, %

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16. Difference between = and ==?

- = → Assignment operator

- == → Comparison operator

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17. What are relational operators?

<, >, <=, >=, ==, !=

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18. What are logical operators?

&&, ||, !

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19. What is the modulus operator (%)?

Returns the remainder after division.

Example:

10 % 3 = 1

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20. What is type casting?

Converting one data type into another.

Example:

float a = (float)10/3;

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21. What is implicit type conversion?

Automatic conversion by the compiler.

Example:

int a = 10;

float b = a;

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22. What is explicit type conversion?

Conversion performed by the programmer.

Example:

float b = (float)a;

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23. What are comments in C?

Used to explain code.

// Single line

/* Multi-line */

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24. What are escape sequences?

Special characters beginning with \.

Examples:

- \n
- \t
- \b

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25. What is operator precedence?

The order in which operators are evaluated.

Example:

2 + 3 * 4 = 14

Medium

26. Difference between for, while, and do-while loops?

Loop	Condition Check
for	Before execution
while	Before execution
do-while	After execution

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27. What is a nested loop?

A loop inside another loop.

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28. Difference between break and continue?

- break → Terminates loop.
- continue → Skips current iteration.

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29. What is a function?

A block of code that performs a specific task.

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30. What are function prototypes?

Declarations that tell the compiler about a function before its use.

Example:

int add(int,int);

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31. Difference between actual and formal parameters?

- Actual: Values passed during function call.
- Formal: Variables receiving those values.

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32. What is recursion?

A function calling itself.

Example:

fact(n)=n*fact(n-1)

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33. What are arrays?

Collection of similar data types stored in contiguous memory.

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34. Difference between 1D and 2D arrays?

- 1D → Single row
- 2D → Rows and columns

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35. What is a string in C?

An array of characters ending with \0.

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36. What is the null character (\0)?

It marks the end of a string.

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37. How are strings stored in memory?

As consecutive characters followed by \0.

Example:

char str[]="CAT";

Memory:

C A T \0

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38. Difference between gets() and fgets()?

- gets() is unsafe.
- fgets() is safer because it limits input size.

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39. What is a pointer?

A variable that stores the address of another variable.

Example:

int *p;

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40. Why are pointers used?

- Dynamic memory allocation
- Efficient memory access
- Pass by reference

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41. What is a NULL pointer?

A pointer that points to no valid memory location.

int *p = NULL;

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42. Difference between *p and p?

- p → Address
- *p → Value at that address

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43. What is pointer arithmetic?

Arithmetic operations on addresses.

Example:

p++;

Moves pointer to the next memory location.

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44. Difference between call by value and call by reference?

Call by Value	Call by Reference
Copy passed	Address passed
Original value unchanged	Original value can change

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45. What are local variables?

Variables declared inside a function and accessible only within it.

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46. What are global variables?

Variables declared outside all functions and accessible throughout the program.

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47. What are storage classes?

They define scope, lifetime, and visibility of variables.

Examples:

- auto
- static
- extern
- register

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48. What is the static keyword?

Preserves a variable's value between function calls.

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49. What is the extern keyword?

Used to access a global variable defined in another file.

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50. What are command-line arguments (argc and argv)?

Used to pass values when running a program.

int main(int argc, char *argv[])

- argc → Number of arguments
- argv → Array of arguments

Example:

./prog hello

argc = 2

argv[1] = hello
